Chapter 6. Differential Forms

Multilinear algebra, alternating tensors and the wedge product, differential forms, the operator d, and pullbacks

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Overview

Differentiation and integration have been treated in generality; the third topic is "vector integral calculus", whose major theorems carry the names of Green, Gauss and Stokes. For k-manifolds in $\mathbb{R}^n$ the concepts of linear algebra and vector calculus are no longer adequate; one needs *multilinear algebra*. Sections 26–28 develop it using only the linear algebra of Chapter 1. The rest of the chapter combines it with differentiation to define *differential forms* in $\mathbb{R}^n$ and their operators, which replace the vector and scalar fields and the operators grad, curl and div of $\mathbb{R}^3$. Integration and Stokes' theorem follow in Chapter 7.

§26. Multilinear Algebra

Motivation. The integrand over a k-dimensional surface must assign a number to each infinitesimal parallelepiped, that is, to k tangent vectors, linearly in each. This is a multilinear function, a tensor; this section develops their algebra.

26.1 Tensors

Definition 26.1 (k-tensor)

Let V be a vector space and $V^k = V \times \cdots \times V$ the set of k-tuples $(\mathbf{v}_1, \ldots, \mathbf{v}_k)$ of vectors of V. A function $f : V^k \to \mathbb{R}$ is **linear in the ith variable** if, for fixed $\mathbf{v}_j$ $(j \neq i)$, the function $T(\mathbf{v}) = f(\mathbf{v}_1, \ldots, \mathbf{v}_{i-1}, \mathbf{v}, \mathbf{v}_{i+1}, \ldots, \mathbf{v}_k)$ is linear. f is **multilinear** if it is linear in the ith variable for each i; it is then called a **k-tensor**, or a **tensor of order k**, on V. The set of k-tensors on V is $\mathcal{L}^k(V)$. $\mathcal{L}^1(V)$ is the set of linear maps $V \to \mathbb{R}$, the **dual space** V^* of V.

Theorem 26.1

The set of all k-tensors on V constitutes a vector space if we define

$$(f+g)(\mathbf{v}_1, \ldots, \mathbf{v}_k) = f(\mathbf{v}_1, \ldots, \mathbf{v}_k) + g(\mathbf{v}_1, \ldots, \mathbf{v}_k), \qquad (cf)(\mathbf{v}_1, \ldots, \mathbf{v}_k) = c(f(\mathbf{v}_1, \ldots, \mathbf{v}_k)).$$

Proof. Sums and scalar multiples of functions linear in each variable are linear in each variable; the vector space axioms hold because they hold for real numbers pointwise. The zero tensor is the function whose value on every k-tuple is zero. ■

Lemma 26.2

Let $\mathbf{a}_1, \ldots, \mathbf{a}_n$ be a basis for V. If $f, g : V^k \to \mathbb{R}$ are k-tensors on V, and if $f(\mathbf{a}_{i_1}, \ldots, \mathbf{a}_{i_k}) = g(\mathbf{a}_{i_1}, \ldots, \mathbf{a}_{i_k})$ for every k-tuple $I = (i_1, \ldots, i_k)$ of integers from the set $\{1, \ldots, n\}$, then $f = g$.

(The integers $i_1, \ldots, i_k$ need not be distinct or in any order.)

Proof. Write $\mathbf{v}_i = \sum_{j=1}^n c_{ij}\mathbf{a}_j$. Expanding by multilinearity, one variable at a time,

$$f(\mathbf{v}_1, \ldots, \mathbf{v}_k) = \sum_{1 \le j_1, \ldots, j_k \le n} c_{1j_1} c_{2j_2} \cdots c_{kj_k}\, f(\mathbf{a}_{j_1}, \ldots, \mathbf{a}_{j_k}),$$

and likewise for g. ■

Theorem 26.3

Let V be a vector space with basis $\mathbf{a}_1, \dots, \mathbf{a}_n$. Let $I = (i_1, \dots, i_k)$ be a k-tuple of integers from the set $\{1, \dots, n\}$. There is a unique k-tensor ϕ_I on V such that, for every k-tuple $J = (j_1, \dots, j_k)$ from the set $\{1, \dots, n\}$,

$$\phi_I(\mathbf{a}_{j_1}, \dots, \mathbf{a}_{j_k}) = \begin{cases} 0 & \text{if } I \neq J, \\ 1 & \text{if } I = J. \end{cases} \qquad (*)$$

The tensors ϕ_I form a basis for $\mathcal{L}^k(V)$.

The ϕ_I are the **elementary k-tensors** on V corresponding to the basis $\mathbf{a}_1, \dots, \mathbf{a}_n$. There are n^k of them, so $\dim \mathcal{L}^k(V) = n^k$. For $k = 1$, the basis $\phi_1, \dots, \phi_n$ of V^* is the basis **dual** to $\mathbf{a}_1, \dots, \mathbf{a}_n$.

Proof. Uniqueness follows from Lemma 26.2.

Existence: for $k = 1$, define the linear map ϕ_i by $\phi_i(\mathbf{a}_j) = 0$ $(i \neq j)$, 1 $(i = j)$ (Theorem 1.4). For $k > 1$ set

$$\phi_I(\mathbf{v}_1, \dots, \mathbf{v}_k) = [\phi_{i_1}(\mathbf{v}_1)] \cdot [\phi_{i_2}(\mathbf{v}_2)] \cdots [\phi_{i_k}(\mathbf{v}_k)];$$

it is multilinear and has the values $(*)$. Given a k-tensor f, let $d_I = f(\mathbf{a}_{i_1}, \dots, \mathbf{a}_{i_k})$ and $g = \sum_J d_J \phi_J$ (sum over all k-tuples J). Then $g(\mathbf{a}_{i_1}, \dots, \mathbf{a}_{i_k}) = d_I$, so $f = g$ by Lemma 26.2; the representation is unique by the same lemma. ■

Thus a k-tensor may be defined by specifying its values arbitrarily on k-tuples of basis elements.

Example 26.1

Let $V = \mathbb{R}^n$ with the usual basis and dual basis $\phi_1, \dots, \phi_n$. Then $\phi_i(\mathbf{x}) = \phi_i(x_1\mathbf{e}_1 + \cdots + x_n\mathbf{e}_n) = x_i$: ϕ_i is projection onto the ith coordinate. More generally $\phi_I(\mathbf{x}_1, \dots, \mathbf{x}_k) = \phi_{i_1}(\mathbf{x}_1) \cdots \phi_{i_k}(\mathbf{x}_k)$; writing $X = [\mathbf{x}_1 \cdots \mathbf{x}_k]$ with entries x_{ij},

$$\phi_I(\mathbf{x}_1, \dots, \mathbf{x}_k) = x_{i_1 1} x_{i_2 2} \cdots x_{i_k k},$$

a *monomial* in the components of the $\mathbf{x}_j$. The general k-tensor on $\mathbb{R}^n$ is a linear combination of such monomials: the general 1-tensor is $f(\mathbf{x}) = d_1 x_1 + \cdots + d_n x_n$, the general 2-tensor is $g(\mathbf{x}, \mathbf{y}) = \sum_{i,j=1}^n d_{ij} x_i y_j$, and so on.

26.2 The tensor product

Definition 26.2 (tensor product)

Let f be a k-tensor and g an l-tensor on V. The $(k + l)$-tensor $f \otimes g$ is

$$(f \otimes g)(\mathbf{v}_1, \dots, \mathbf{v}_{k+l}) = f(\mathbf{v}_1, \dots, \mathbf{v}_k) \cdot g(\mathbf{v}_{k+1}, \dots, \mathbf{v}_{k+l}).$$

It is multilinear, and is the **tensor product** of f and g.

Theorem 26.4

Let f, g, h be tensors on V. Then the following properties hold:

(1) (Associativity) $f \otimes (g \otimes h) = (f \otimes g) \otimes h$.

(2) (Homogeneity) $(cf) \otimes g = c(f \otimes g) = f \otimes (cg)$.

(3) (Distributivity) Suppose f and g have the same order. Then $(f + g) \otimes h = f \otimes h + g \otimes h$

and $h \otimes (f+g) = h \otimes f + h \otimes g$.

(4) Given a basis $\mathbf{a}_1, \ldots, \mathbf{a}_n$ for V, the corresponding elementary tensors ϕ_I satisfy $\phi_I = \phi_{i_1} \otimes \phi_{i_2} \otimes \cdots \otimes \phi_{i_k}$, where $I = (i_1, \ldots, i_k)$.

No parentheses are needed in (4) by associativity; nothing is said about commutativity, which almost never holds.

Proof. Straightforward. For instance, if f, g, h have orders k, l, m, both $(f \otimes (g \otimes h))(\mathbf{v}_1, \ldots, \mathbf{v}_{k+l+m})$ and $((f \otimes g) \otimes h)(\mathbf{v}_1, \ldots, \mathbf{v}_{k+l+m})$ equal $f(\mathbf{v}_1, \ldots, \mathbf{v}_k) \cdot g(\mathbf{v}_{k+1}, \ldots, \mathbf{v}_{k+l}) \cdot h(\mathbf{v}_{k+l+1}, \ldots, \mathbf{v}_{k+l+m})$. Property (4) is the formula in the proof of Theorem 26.3. ■

26.3 The action of a linear transformation

Definition 26.3 (dual transformation)

Let $T : V \to W$ be linear. The **dual transformation** $T^* : \mathcal{L}^k(W) \to \mathcal{L}^k(V)$ (in the opposite direction) is

$$(T^* f)(\mathbf{v}_1, \ldots, \mathbf{v}_k) = f(T(\mathbf{v}_1), \ldots, T(\mathbf{v}_k)),$$

i.e. $T^* f$ is the composite of $T \times \cdots \times T : V^k \to W^k$ with $f : W^k \to \mathbb{R}$.

$T^* f$ is multilinear since T is linear and f multilinear.

Theorem 26.5

Let $T : V \to W$ be a linear transformation; let $T^* : \mathcal{L}^k(W) \to \mathcal{L}^k(V)$ be the dual transformation. Then:
(1) T^* is linear.
(2) $T^*(f \otimes g) = T^* f \otimes T^* g$.
(3) If $S : W \to X$ is a linear transformation, then $(S \circ T)^* f = T^*(S^* f)$.

Proof. Straightforward; e.g. for (1),

$$(T^*(af + bg))(\mathbf{v}_1, \ldots, \mathbf{v}_k) = (af + bg)(T(\mathbf{v}_1), \ldots, T(\mathbf{v}_k)) = aT^* f(\mathbf{v}_1, \ldots, \mathbf{v}_k) + bT^* g(\mathbf{v}_1, \ldots, \mathbf{v}_k).$$

For (2), both sides evaluated at $(\mathbf{v}_1, \ldots, \mathbf{v}_{k+l})$ give $f(T\mathbf{v}_1, \ldots, T\mathbf{v}_k)\, g(T\mathbf{v}_{k+1}, \ldots, T\mathbf{v}_{k+l})$. For (3), both sides give $f(S(T(\mathbf{v}_1)), \ldots, S(T(\mathbf{v}_k)))$. ■

Property (3) says the diagram $V \xrightarrow{T} W \xrightarrow{S} X$ induces $\mathcal{L}^k(X) \xrightarrow{S^*} \mathcal{L}^k(W) \xrightarrow{T^*} \mathcal{L}^k(V)$ with $(S \circ T)^* = T^* \circ S^*$.

§27. Alternating Tensors

Motivation. Orientation matters in integration: reversing the order of two vectors should reverse the sign, and degenerate parallelepipeds should get zero. Alternating tensors have exactly these properties, and the determinant is the model example.

27.1 Permutations

Definition 27.1 (permutation, symmetric group)

Let $k \geq 2$. A **permutation** of $\{1, \ldots, k\}$ is a one-to-one map σ of this set onto itself; the set of them is S_k. With composition and inverses, S_k is a group, the **symmetric group** (**permutation group**); it has $k!$ elements.

Definition 27.2 (elementary permutation)

For $1 \leq i < k$, let $e_i \in S_k$ be defined by $e_i(j) = j$ for $j \neq i, i+1$, $e_i(i) = i+1$, $e_i(i+1) = i$. e_i is an **elementary permutation**; $e_i \circ e_i$ is the identity.

Lemma 27.1

If $\sigma \in S_k$, then σ equals a composite of elementary permutations.

Proof. Say σ *fixes the first* i *integers* if $\sigma(j) = j$ for $1 \leq j \leq i$ $(0 \leq i \leq k)$. If $i = k$, σ is the identity, which equals $e_j \circ e_j$. We show: if σ fixes the first $i-1$ integers $(0 < i \leq k)$, then $\sigma = \pi \circ \sigma'$ with π a composite of elementary permutations and σ' fixing the first i integers; induction completes the proof. Since σ fixes $1, \ldots, i-1$ and is one-to-one, $\sigma(i) = l \geq i$. If $l = i$, take $\sigma' = \sigma$, $\pi =$ identity. If $l > i$, let $\sigma' = e_i \circ \cdots \circ e_{l-1} \circ \sigma$. Then σ' fixes $1, \ldots, i-1$ (as σ and $e_i, \ldots, e_{l-1}$ do), and $\sigma'(i) = e_i(\cdots(e_{l-1}(l))\cdots) = i$. And $\sigma = e_{l-1} \circ \cdots \circ e_i \circ \sigma'$. ■

Definition 27.3 (sign of a permutation)

Let $\sigma \in S_k$. A pair $i < j$ with $\sigma(i) > \sigma(j)$ is an **inversion** in σ. The **sign** $\operatorname{sgn}\sigma$ is -1 if the number of inversions is odd and $+1$ if it is even; σ is **odd** or **even** accordingly.

Lemma 27.2

Let $\sigma, \tau \in S_k$.

(a) If σ equals a composite of m elementary permutations, then $\operatorname{sgn}\sigma = (-1)^m$.

(b) $\operatorname{sgn}(\sigma \circ \tau) = (\operatorname{sgn}\sigma) \cdot (\operatorname{sgn}\tau)$.

(c) $\operatorname{sgn}\sigma^{-1} = \operatorname{sgn}\sigma$.

(d) If $p \neq q$, and if τ is the permutation that exchanges p and q and leaves all other integers fixed, then $\operatorname{sgn}\tau = -1$.

Proof.

Step 1. For any σ, $\operatorname{sgn}(\sigma \circ e_l) = -\operatorname{sgn}\sigma$. Write the values of σ in order: $(\sigma(1), \ldots, \sigma(l), \sigma(l+1), \ldots, \sigma(k))$ $(*)$; for $\tau = \sigma \circ e_l$ the sequence is $(\sigma(1), \ldots, \sigma(l+1), \sigma(l), \ldots, \sigma(k))$ $(**)$. Inversions are pairs of entries appearing in reverse natural order. For $p \neq q$: if neither is l or $l+1$, $\sigma(p), \sigma(q)$ occupy the same slots in both sequences; if one is l or $l+1$ and the other is not, they are in the same relative order in both. Only the pair $\sigma(l), \sigma(l+1)$ changes: it is an inversion in exactly one of the two sequences. So $(**)$ has one more or one fewer inversion than $(*)$.

Step 2. The identity has sign $+1$, and composing with m elementary permutations changes the sign m times: (a). Writing σ, τ as composites of m, n elementary permutations, (b) follows from $(-1)^{m+n} = (-1)^m(-1)^n$.

(c) follows from (b) since $\sigma^{-1} \circ \sigma$ is the identity.

(d) Count inversions: for $p < q$, $q = p + l$, the values of τ in order are $(1, \dots, p-1, q, p+1, \dots, p+l-1, p, p+l+1, \dots, k)$. The pairs $\{q, p+1\}, \dots, \{q, p+l-1\}$ and $\{p+1, p\}, \dots, \{p+l-1, p\}$ are inversions, and so is $\{q, p\}$: $2l-1$ in all, an odd number. ■

27.2 Alternating tensors

Definition 27.4 (symmetric and alternating tensors)

Let f be a k-tensor on V and $\sigma \in S_k$. Define the k-tensor f^σ by

$$f^\sigma(\mathbf{v}_1, \dots, \mathbf{v}_k) = f(\mathbf{v}_{\sigma(1)}, \dots, \mathbf{v}_{\sigma(k)}).$$

f is **symmetric** if $f^e = f$ for each elementary permutation e, and **alternating** if $f^e = -f$ for every elementary permutation e; i.e. f is alternating if

$$f(\mathbf{v}_1, \dots, \mathbf{v}_{i+1}, \mathbf{v}_i, \dots, \mathbf{v}_k) = -f(\mathbf{v}_1, \dots, \mathbf{v}_i, \mathbf{v}_{i+1}, \dots, \mathbf{v}_k) \quad \text{for all } i.$$

The set of alternating k-tensors on V is $\mathcal{A}^k(V)$, a linear subspace of $\mathcal{L}^k(V)$. By convention, $\mathcal{A}^1(V) = \mathcal{L}^1(V)$.

We are primarily interested in alternating tensors.

Example 27.1

Elementary tensors of order $k > 1$ are not alternating, but certain combinations are. With $V = \mathbb{R}^n$ and the dual basis ϕ_i, $f = \phi_{i,j} - \phi_{j,i}$ satisfies

$$f(\mathbf{x}, \mathbf{y}) = x_i y_j - x_j y_i = \det \begin{bmatrix} x_i & y_i \\ x_j & y_j \end{bmatrix},$$

so $f(\mathbf{y}, \mathbf{x}) = -f(\mathbf{x}, \mathbf{y})$. Similarly $g(\mathbf{x}, \mathbf{y}, \mathbf{z}) = \det \begin{bmatrix} x_i & y_i & z_i \\ x_j & y_j & z_j \\ x_k & y_k & z_k \end{bmatrix}$ is an alternating 3-tensor on $\mathbb{R}^n$; $g = \phi_{i,j,k} + \phi_{j,k,i} + \phi_{k,i,j} - \phi_{j,i,k} - \phi_{i,k,j} - \phi_{k,j,i}$. Alternating tensors and determinants are intimately related.

Lemma 27.3

Let f be a k-tensor on V; let $\sigma, \tau \in S_k$.

(a) The transformation $f \to f^\sigma$ is a linear transformation of $\mathcal{L}^k(V)$ to $\mathcal{L}^k(V)$. It has the property that for all σ, τ, $(f^\sigma)^\tau = f^{\tau \circ \sigma}$.

(b) The tensor f is alternating if and only if $f^\sigma = (\operatorname{sgn} \sigma) f$ for all σ. If f is alternating and if $\mathbf{v}_p = \mathbf{v}_q$ with $p \neq q$, then $f(\mathbf{v}_1, \dots, \mathbf{v}_k) = 0$.

Proof. (a) Linearity says $(af + bg)^\sigma = af^\sigma + bg^\sigma$. With $\mathbf{w}_i = \mathbf{v}_{\tau(i)}$,

$$(f^\sigma)^\tau(\mathbf{v}_1, \dots, \mathbf{v}_k) = f^\sigma(\mathbf{w}_1, \dots, \mathbf{w}_k) = f(\mathbf{w}_{\sigma(1)}, \dots, \mathbf{w}_{\sigma(k)}) = f(\mathbf{v}_{\tau(\sigma(1))}, \dots, \mathbf{v}_{\tau(\sigma(k))}) = f^{\tau\circ\sigma}(\mathbf{v}_1, \dots, \mathbf{v}_k).$$

(b) If $f^\sigma = (\operatorname{sgn} \sigma) f$ for all σ, then $f^e = -f$ for elementary e. Conversely, write $\sigma = \sigma_1 \circ \cdots \circ \sigma_m$ with σ_i elementary; by (a), $f^\sigma = ((\cdots (f^{\sigma_m}) \cdots)^{\sigma_2})^{\sigma_1} = (-1)^m f = (\operatorname{sgn} \sigma) f$. If $\mathbf{v}_p = \mathbf{v}_q$ $(p \neq q)$ and τ exchanges p and q, then $f^\tau(\mathbf{v}_1, \dots, \mathbf{v}_k) = f(\mathbf{v}_1, \dots, \mathbf{v}_k)$, while $f^\tau = -f$ since $\operatorname{sgn} \tau = -1$; so $f(\mathbf{v}_1, \dots, \mathbf{v}_k) = 0$. ■

For $k = 1$ nothing is to be done. For $k > n$, $\mathcal{A}^k(V)$ is trivial: some basis element appears twice in each k-tuple of basis elements, so an alternating f vanishes on all of them (Lemma 26.2). For $1 < k \leq n$, an alternating tensor is determined by its values on k-tuples of basis elements with indices in ascending order, which may be specified arbitrarily.

Lemma 27.4

Let $\mathbf{a}_1, \ldots, \mathbf{a}_n$ be a basis for V. If f, g are alternating k-tensors on V, and if $f(\mathbf{a}_{i_1}, \ldots, \mathbf{a}_{i_k}) = g(\mathbf{a}_{i_1}, \ldots, \mathbf{a}_{i_k})$ for every ascending k-tuple of integers $I = (i_1, \ldots, i_k)$ from the set $\{1, \ldots, n\}$, then $f = g$.

Proof. By Lemma 26.2 it suffices to show f, g agree on an arbitrary $(\mathbf{a}_{j_1}, \ldots, \mathbf{a}_{j_k})$. If two indices j_p, j_q coincide, both values are 0 (Lemma 27.3). Otherwise let σ be the permutation such that $I = (j_{\sigma(1)}, \ldots, j_{\sigma(k)})$ is ascending. Then $f(\mathbf{a}_{i_1}, \ldots, \mathbf{a}_{i_k}) = f^\sigma(\mathbf{a}_{j_1}, \ldots, \mathbf{a}_{j_k}) = (\operatorname{sgn}\sigma) f(\mathbf{a}_{j_1}, \ldots, \mathbf{a}_{j_k})$, and similarly for g. Since f, g agree on the ascending tuple, they agree on $(\mathbf{a}_{j_1}, \ldots, \mathbf{a}_{j_k})$. ■

Theorem 27.5

Let V be a vector space with basis $\mathbf{a}_1, \ldots, \mathbf{a}_n$. Let $I = (i_1, \ldots, i_k)$ be an ascending k-tuple from the set $\{1, \ldots, n\}$. There is a unique alternating k-tensor ψ_I on V such that for every ascending k-tuple $J = (j_1, \ldots, j_k)$ from the set $\{1, \ldots, n\}$,

$$\psi_I(\mathbf{a}_{j_1}, \ldots, \mathbf{a}_{j_k}) = \begin{cases} 0 & \text{if } I \neq J, \\ 1 & \text{if } I = J. \end{cases}$$

The tensors ψ_I form a basis for $\mathcal{A}^k(V)$. The tensor ψ_I in fact satisfies the formula

$$\psi_I = \sum_\sigma (\operatorname{sgn}\sigma)(\phi_I)^\sigma,$$

where the summation extends over all $\sigma \in S_k$.

The ψ_I are the **elementary alternating k-tensors** on V corresponding to the basis $\mathbf{a}_1, \ldots, \mathbf{a}_n$.

Proof. Uniqueness follows from Lemma 27.4. Define ψ_I by the formula. It is alternating: for $\tau \in S_k$,

$$(\psi_I)^\tau = \sum_\sigma (\operatorname{sgn}\sigma)((\phi_I)^\sigma)^\tau = \sum_\sigma (\operatorname{sgn}\sigma)(\phi_I)^{\tau\circ\sigma} = (\operatorname{sgn}\tau)\sum_\sigma (\operatorname{sgn}(\tau\circ\sigma))(\phi_I)^{\tau\circ\sigma} = (\operatorname{sgn}\tau)\psi_I,$$

since $\tau \circ \sigma$ ranges over S_k as σ does. Its values: $\psi_I(\mathbf{a}_{j_1}, \ldots, \mathbf{a}_{j_k}) = \sum_\sigma (\operatorname{sgn}\sigma)\phi_I(\mathbf{a}_{j_{\sigma(1)}}, \ldots, \mathbf{a}_{j_{\sigma(k)}})$. At most one term is nonzero, the one with $I = (j_{\sigma(1)}, \ldots, j_{\sigma(k)})$; as I and J are both ascending this happens only if $I = J$ and σ is the identity, giving value 1.

The ψ_I form a basis: given alternating f, let $d_I = f(\mathbf{a}_{i_1}, \ldots, \mathbf{a}_{i_k})$ for ascending I, and $g = \sum_{[J]} d_J \psi_J$ (sum over ascending J). Then f and g agree on ascending k-tuples of basis elements, so $f = g$ by Lemma 27.4; uniqueness of the representation follows from the same lemma. ■

Thus, once a basis is chosen, each alternating k-tensor can be written uniquely as $f = \sum_{[J]} d_J \psi_J$; the d_J are the **components** of f relative to $\{\psi_J\}$. Since each k-element subset of $\{1, \ldots, n\}$ gives exactly one ascending k-tuple,

$$\dim \mathcal{A}^k(V) = \binom{n}{k} = \frac{n!}{k!(n-k)!}.$$

Theorem 27.6

Let $T : V \to W$ be a linear transformation. If f is an alternating tensor on W, then T^*f is an alternating tensor on V.

Proof [Supplement]. $(T^*f)(\dots, \mathbf{v}_{i+1}, \mathbf{v}_i, \dots) = f(\dots, T\mathbf{v}_{i+1}, T\mathbf{v}_i, \dots) = -f(\dots, T\mathbf{v}_i, T\mathbf{v}_{i+1}, \dots) = -(T^*f)(\dots, \mathbf{v}_i, \mathbf{v}_{i+1}, \dots)$. ■

27.3 Determinants

We now (at long last) construct the determinant for matrices larger than 3 by 3 (cf. §2).

Definition 27.5 (determinant)

Let $\phi_1, \dots, \phi_n$ be the basis of $\mathcal{L}^1(\mathbb{R}^n)$ dual to the usual basis. $\mathcal{A}^n(\mathbb{R}^n)$ has dimension 1; its unique elementary alternating n-tensor is $\psi_{1,\dots,n}$. For an n by n matrix $X = [\mathbf{x}_1 \ \cdots \ \mathbf{x}_n]$ define

$$\det X = \psi_{1,\dots,n}(\mathbf{x}_1, \dots, \mathbf{x}_n).$$

This satisfies the axioms of §2. Let $g(X) = \psi_I(\mathbf{x}_1, \dots, \mathbf{x}_n)$ with $I = (1, \dots, n)$; g is multilinear and alternating in the *columns* of X. So $f(A) = g(A^{\text{tr}})$ is multilinear and alternating in the rows of A, and $f(I_n) = g(I_n) = \psi_I(\mathbf{e}_1, \dots, \mathbf{e}_n) = 1$; hence f satisfies the axioms. By Theorem 2.11, $f(A) = f(A^{\text{tr}})$, so $f(A) = g(A)$ and g satisfies the axioms too. Theorem 27.5 gives the formula

$$\det X = \sum_\sigma (\operatorname{sgn}\sigma)\phi_I(\mathbf{x}_{\sigma(1)}, \dots, \mathbf{x}_{\sigma(n)}) = \sum_\sigma (\operatorname{sgn}\sigma) x_{1,\sigma(1)} \cdot x_{2,\sigma(2)} \cdots x_{n,\sigma(n)},$$

sometimes used as the definition of the determinant.

Theorem 27.7

Let ψ_I be an elementary alternating tensor on $\mathbb{R}^n$ corresponding to the usual basis for $\mathbb{R}^n$, where $I = (i_1, \dots, i_k)$. Given vectors $\mathbf{x}_1, \dots, \mathbf{x}_k$ of $\mathbb{R}^n$, let X be the matrix $X = [\mathbf{x}_1 \ \cdots \ \mathbf{x}_k]$. Then

$$\psi_I(\mathbf{x}_1, \dots, \mathbf{x}_k) = \det X_I,$$

where X_I denotes the matrix whose successive rows are rows $i_1, \dots, i_k$ of X.

Proof. $\psi_I(\mathbf{x}_1, \dots, \mathbf{x}_k) = \sum_\sigma (\operatorname{sgn}\sigma)\phi_I(\mathbf{x}_{\sigma(1)}, \dots, \mathbf{x}_{\sigma(k)}) = \sum_\sigma (\operatorname{sgn}\sigma) x_{i_1,\sigma(1)} \cdot x_{i_2,\sigma(2)} \cdots x_{i_k,\sigma(k)}$, which is the formula for $\det X_I$. ■

Example 27.2

The elementary alternating 3-tensors on $\mathbb{R}^4$ are

$$\psi_{i,j,k}(\mathbf{x}, \mathbf{y}, \mathbf{z}) = \det \begin{bmatrix} x_i & y_i & z_i \\ x_j & y_j & z_j \\ x_k & y_k & z_k \end{bmatrix},$$

for $(i, j, k) = (1, 2, 3), (1, 2, 4), (1, 3, 4), (2, 3, 4)$; the general element of $\mathcal{A}^3(\mathbb{R}^4)$ is a linear combination of these four.

Notation

Multilinear algebra has the *exterior product* construction $\Lambda^k(W)$, a quotient of the k-fold tensor product of W. For finite-dimensional V, $\Lambda^k(V^*)$ is naturally isomorphic to the space of alternating k-tensors on V, which some authors therefore denote $\Lambda^k(V^*)$, others $\Lambda^k(V)$. To avoid this confusion we use the neutral notation $\mathcal{A}^k(V)$.

§28. The Wedge Product

Motivation. We need to multiply alternating tensors so that $dx \wedge dy$ represents oriented area. The wedge product is the alternating version of the tensor product and makes the alternating tensors into an algebra with basis dx_I.

$f \otimes g$ is almost never alternating, even if f and g are; something else is needed. The actual definition matters less than its properties:

Theorem 28.1

Let V be a vector space. There is an operation that assigns, to each $f \in \mathcal{A}^k(V)$ and each $g \in \mathcal{A}^l(V)$, an element $f \wedge g \in \mathcal{A}^{k+l}(V)$, such that the following properties hold:

(1) (Associativity) $f \wedge (g \wedge h) = (f \wedge g) \wedge h$.

(2) (Homogeneity) $(cf) \wedge g = c(f \wedge g) = f \wedge (cg)$.

(3) (Distributivity) If f and g have the same order, $(f + g) \wedge h = f \wedge h + g \wedge h$ and $h \wedge (f + g) = h \wedge f + h \wedge g$.

(4) (Anticommutativity) If f and g have orders k and l, respectively, then $g \wedge f = (-1)^{kl} f \wedge g$.

(5) Given a basis $\mathbf{a}_1, \ldots, \mathbf{a}_n$ for V, let ϕ_i denote the dual basis for V^*, and let ψ_I denote the corresponding elementary alternating tensors. If $I = (i_1, \ldots, i_k)$ is an ascending k-tuple of integers from the set $\{1, \ldots, n\}$, then $\psi_I = \phi_{i_1} \wedge \phi_{i_2} \wedge \cdots \wedge \phi_{i_k}$.

These five properties characterize the product $\wedge$ uniquely for finite-dimensional spaces V. Furthermore, it has the following additional property:

(6) If $T : V \to W$ is a linear transformation, and if f and g are alternating tensors on W, then $T^*(f \wedge g) = T^* f \wedge T^* g$.

$f \wedge g$ is the **wedge product** of f and g. By (4), $f \wedge f = 0$ for an alternating tensor f of odd order.

Proof.

Step 1. For a k-tensor F on V (not necessarily alternating) define $A : \mathcal{L}^k(V) \to \mathcal{L}^k(V)$ by

$$AF = \sum_\sigma (\operatorname{sgn}\sigma) F^\sigma,$$

summed over $\sigma \in S_k$. (Some authors include a factor $1/k!$.) Thus $\psi_I = A\phi_I$. A has the properties:

(i) A is linear;

(ii) AF is alternating;

(iii) if F is already alternating, $AF = (k!)F$. Indeed (i) holds since $F \mapsto F^\sigma$ is linear;

(ii) by the computation $(AF)^\tau = \sum_\sigma (\operatorname{sgn}\sigma) F^{\tau\circ\sigma} = (\operatorname{sgn}\tau) \sum_\sigma (\operatorname{sgn}\tau\circ\sigma) F^{\tau\circ\sigma} = (\operatorname{sgn}\tau) AF$;

(iii) since $F^\sigma = (\operatorname{sgn}\sigma) F$ gives $AF = \sum_\sigma (\operatorname{sgn}\sigma)^2 F = (k!)F$.

Step 2. For $f \in \mathcal{A}^k(V)$, $g \in \mathcal{A}^l(V)$ define

$$f \wedge g = \frac{1}{k!\,l!} A(f \otimes g),$$

an alternating tensor of order $k+l$. *Motivation for* $1/k!l!$: in $(f\wedge g)(\mathbf{v}_1, \dots, \mathbf{v}_{k+l}) = \frac{1}{k!l!}\sum_\sigma (\operatorname{sgn}\sigma) f(\mathbf{v}_{\sigma(1)}, \dots, \mathbf{v}_{\sigma(k)})$ $g(\mathbf{v}_{\sigma(k+1)}, \dots, \mathbf{v}_{\sigma(k+l)})$, permuting $\mathbf{v}_{\sigma(1)}, \dots, \mathbf{v}_{\sigma(k)}$ among themselves and $\mathbf{v}_{\sigma(k+1)}, \dots, \mathbf{v}_{\sigma(k+l)}$ among themselves gives $k!l!$ terms with the same value (since f, g are alternating, the sign changes cancel); dividing by $k!l!$ removes this redundancy.

Step 3. Homogeneity: $(cf) \wedge g = A((cf)\otimes g)/k!l! = A(c(f\otimes g))/k!l! = cA(f\otimes g)/k!l! = c(f\wedge g)$; similarly for the other part. Distributivity follows from distributivity of $\otimes$ and linearity of A.

Step 4. More generally, for tensors F, G of orders k, l (not necessarily alternating), $A(F\otimes G) = (-1)^{kl} A(G\otimes F)$. Let $\pi \in S_{k+l}$ with $(\pi(1), \dots, \pi(k+l)) = (k+1, k+2, \dots, k+l, 1, 2, \dots, k)$; counting inversions, $\operatorname{sgn}\pi = (-1)^{kl}$. Since $(G\otimes F)^\pi(\mathbf{v}_1, \dots, \mathbf{v}_{k+l}) = G(\mathbf{v}_{k+1}, \dots, \mathbf{v}_{k+l}) \cdot F(\mathbf{v}_1, \dots, \mathbf{v}_k)$, we have $(G\otimes F)^\pi = F \otimes G$. Hence

$$A(F\otimes G) = \sum_\sigma (\operatorname{sgn}\sigma)((G\otimes F)^\pi)^\sigma = (\operatorname{sgn}\pi)\sum_\sigma (\operatorname{sgn}\sigma\circ\pi)(G\otimes F)^{\sigma\circ\pi} = (\operatorname{sgn}\pi) A(G\otimes F),$$

since $\sigma\circ\pi$ runs over S_{k+l}. Taking $F = f$, $G = g$ gives anticommutativity.

Step 5. If F, G have orders k, l and $AF = 0$, then $A(F\otimes G) = 0$. In $A(F\otimes G)$, group together the terms with the same last factor $G(\mathbf{v}_{\sigma(k+1)}, \dots, \mathbf{v}_{\sigma(k+l)})$. They are

$$(\operatorname{sgn}\sigma)\Big[\sum_\tau (\operatorname{sgn}\tau) F(\mathbf{v}_{\sigma(\tau(1))}, \dots, \mathbf{v}_{\sigma(\tau(k))})\Big] \cdot G(\mathbf{v}_{\sigma(k+1)}, \dots, \mathbf{v}_{\sigma(k+l)}),$$

τ ranging over S_k, and the bracket is $AF(\mathbf{v}_{\sigma(1)}, \dots, \mathbf{v}_{\sigma(k)}) = 0$. Each group cancels.

Step 6. If F is any tensor and h alternating of order m, then $(AF)\wedge h = \frac{1}{m!} A(F\otimes h)$. With F of order k this reads $\frac{1}{k!m!}A((AF)\otimes h) = \frac{1}{m!}A(F\otimes h)$, i.e. (by linearity and distributivity) $A\{[AF - (k!)F]\otimes h\} = 0$. By Step 5 it suffices that $A[AF - (k!)F] = 0$, which follows from property (iii) of A, since AF is alternating: $A(AF) = (k!)AF$.

Step 7. For alternating f, g, h of orders k, l, m: $(f\wedge g)\wedge h = \frac{1}{k!l!m!}A((f\otimes g)\otimes h)$. Indeed with $F = f\otimes g$, $f\wedge g = \frac{1}{k!l!}AF$, so by Step 6, $(f\wedge g)\wedge h = \frac{1}{k!l!}(AF)\wedge h = \frac{1}{k!l!m!}A(F\otimes h)$.

Step 8. Associativity:

$$\begin{aligned}
(k!l!m!)(f\wedge g)\wedge h &= A((f\otimes g)\otimes h) && \text{by Step 7,}\\
&= A(f\otimes(g\otimes h)) && \text{by associativity of } \otimes,\\
&= (-1)^{k(l+m)} A((g\otimes h)\otimes f) && \text{by Step 4,}\\
&= (-1)^{k(l+m)}(l!m!k!)(g\wedge h)\wedge f && \text{by Step 7,}\\
&= (k!l!m!) f\wedge(g\wedge h) && \text{by anticommutativity.}
\end{aligned}$$

Step 9. Property (5). More generally, for 1-tensors $f_1, \dots, f_k$,

$$A(f_1\otimes\cdots\otimes f_k) = f_1\wedge\cdots\wedge f_k. \qquad (*)$$

Then (5) follows since $\psi_I = A\phi_I = A(\phi_{i_1}\otimes\cdots\otimes\phi_{i_k})$. $(*)$ is trivial for $k = 1$; if true for $k-1$, let $F = f_1\otimes\cdots\otimes f_{k-1}$; by Step 6 with $m = 1$, $A(F\otimes f_k) = (AF)\wedge f_k = (f_1\wedge\cdots\wedge f_{k-1})\wedge f_k$.

Step 10. Uniqueness: with only (1)–(5) one can compute wedge products in finite dimensions. Write $f = \sum_{[I]} b_I\psi_I$, $g = \sum_{[J]} c_J\psi_J$; distributivity and homogeneity give $f\wedge g = \sum_{[I]}\sum_{[J]} b_I c_J \psi_I\wedge\psi_J$, and $\psi_I\wedge\psi_J = (\phi_{i_1}\wedge\cdots\wedge\phi_{i_k})\wedge(\phi_{j_1}\wedge\cdots\wedge\phi_{j_l})$. By associativity and the rules $\phi_i\wedge\phi_j = -\phi_j\wedge\phi_i$, $\phi_i\wedge\phi_i = 0$ (from anticommutativity), $\psi_I\wedge\psi_J = 0$ if two indices coincide, and otherwise equals $\operatorname{sgn}\pi$ times ψ_K, where K is the ascending rearrangement of (I, J) and π the permutation effecting it.

Step 11. Property (6). For any tensor F on W, $T^*(F^\sigma) = (T^*F)^\sigma$; since T^* is linear, $T^*(AF) = A(T^*F)$. For alternating f, g of orders k, l on W,

$$T^*(f \wedge g) = \frac{1}{k!l!} T^*(A(f \otimes g)) = \frac{1}{k!l!} A(T^*(f \otimes g)) = \frac{1}{k!l!} A((T^*f) \otimes (T^*g)) = (T^*f) \wedge (T^*g),$$

using Theorem 26.5. ∎

Remark

Some texts define the wedge product with coefficient $1/(k+l)!$ instead of $1/k!l!$. This also gives an associative operation, and all properties of Theorem 28.1 persist except (5), which acquires a factor $k!$ on the right side of the formula for ψ_I. We need only alternating tensors and the properties above.

§29. Tangent Vectors and Differential Forms

Calculus studies vector algebra in $\mathbb{R}^3$, scalar and vector fields, and the operators $\operatorname{grad} f = \nabla f$, $\operatorname{curl} \vec{F} = \nabla \times \vec{F}$, $\operatorname{div} \vec{G} = \nabla \cdot \vec{G}$. Analogously, having studied tensor algebra in $\mathbb{R}^n$, we now introduce tensor fields, especially alternating tensor fields (*differential forms*), and in §30 the *differential operator* d, the analogue of grad, curl and div.

29.1 Tangent vectors and vector fields

Definition 29.1 (tangent vector, tangent space)

Given $\mathbf{x} \in \mathbb{R}^n$, a **tangent vector** to $\mathbb{R}^n$ at $\mathbf{x}$ is a pair $(\mathbf{x}; \mathbf{v})$ with $\mathbf{v} \in \mathbb{R}^n$. The tangent vectors at $\mathbf{x}$ form a vector space, the **tangent space** $\mathcal{T}_{\mathbf{x}}(\mathbb{R}^n)$, under

$$(\mathbf{x}; \mathbf{v}) + (\mathbf{x}; \mathbf{w}) = (\mathbf{x}; \mathbf{v} + \mathbf{w}), \qquad c(\mathbf{x}; \mathbf{v}) = (\mathbf{x}; c\mathbf{v}).$$

We picture $\mathbf{x}$ as a point and $\mathbf{v}$ as an arrow with initial point $\mathbf{x}$; $\mathcal{T}_{\mathbf{x}}(\mathbb{R}^n) = \mathbf{x} \times \mathbb{R}^n$. We do not add $(\mathbf{x}; \mathbf{v}) + (\mathbf{y}; \mathbf{w})$ if $\mathbf{x} \neq \mathbf{y}$.

Definition 29.2 (velocity vector)

Let (a, b) be an open interval and $\gamma : (a, b) \to \mathbb{R}^n$ of class C^r. The **velocity vector** of γ at parameter value t is $(\gamma(t); D\gamma(t))$.

For $\gamma(t) = x(t)\mathbf{e}_1 + y(t)\mathbf{e}_2 + z(t)\mathbf{e}_3$ in $\mathbb{R}^3$, calculus defines the velocity as $D\gamma(t) = \frac{dx}{dt}\mathbf{e}_1 + \frac{dy}{dt}\mathbf{e}_2 + \frac{dz}{dt}\mathbf{e}_3$.

Definition 29.3 (induced transformation α_*)

Let A be open in $\mathbb{R}^k$ or $\mathbb{H}^k$, $\alpha : A \to \mathbb{R}^n$ of class C^r, $\mathbf{x} \in A$, $\mathbf{p} = \alpha(\mathbf{x})$. The linear transformation $\alpha_* : \mathcal{T}_{\mathbf{x}}(\mathbb{R}^k) \to \mathcal{T}_{\mathbf{p}}(\mathbb{R}^n)$ **induced** by α is

$$\alpha_*(\mathbf{x}; \mathbf{v}) = (\mathbf{p}; D\alpha(\mathbf{x}) \cdot \mathbf{v}).$$

By the chain rule, $\alpha_*(\mathbf{x}; \mathbf{v})$ is the velocity vector of the curve $\gamma(t) = \alpha(\mathbf{x} + t\mathbf{v})$ at $t = 0$.

Lemma 29.1

Let A be open in $\mathbb{R}^k$ or $\mathbb{H}^k$; let $\alpha : A \to \mathbb{R}^m$ be of class C^r. Let B be an open set of $\mathbb{R}^m$ or $\mathbb{H}^m$ containing $\alpha(A)$; let $\beta : B \to \mathbb{R}^n$ be of class C^r. Then $(\beta \circ \alpha)_* = \beta_* \circ \alpha_*$.

Proof. With $\mathbf{y} = \alpha(\mathbf{x})$, $\mathbf{z} = \beta(\mathbf{y})$, the chain rule gives $(\beta \circ \alpha)_*(\mathbf{x}; \mathbf{v}) = (\mathbf{z}; D\beta(\mathbf{y}) \cdot D\alpha(\mathbf{x}) \cdot \mathbf{v}) = \beta_*(\mathbf{y}; D\alpha(\mathbf{x}) \cdot \mathbf{v}) = \beta_*(\alpha_*(\mathbf{x}; \mathbf{v}))$. ■

Definition 29.4 (tangent vector field)

If A is open in $\mathbb{R}^n$, a **tangent vector field** in A is a continuous $F : A \to \mathbb{R}^n \times \mathbb{R}^n$ with $F(\mathbf{x}) \in \mathcal{T}_{\mathbf{x}}(\mathbb{R}^n)$ for each $\mathbf{x}$; so $F(\mathbf{x}) = (\mathbf{x}; f(\mathbf{x}))$ for some $f : A \to \mathbb{R}^n$. If f is of class C^r, F is of class C^r.

Definition 29.5 (tangent space to a manifold)

Let M be a k-manifold of class C^r in $\mathbb{R}^n$, $\mathbf{p} \in M$, and $\alpha : U \to V$ a coordinate patch about $\mathbf{p}$ with $\alpha(\mathbf{x}) = \mathbf{p}$. The **tangent space to M at $\mathbf{p}$** is $\mathcal{T}_{\mathbf{p}}(M) = \alpha_*(\mathcal{T}_{\mathbf{x}}(\mathbb{R}^k))$, the set of vectors $\alpha_*(\mathbf{x}; \mathbf{v})$, $\mathbf{v} \in \mathbb{R}^k$.

$\mathcal{T}_{\mathbf{p}}(M)$ is a linear subspace of $\mathcal{T}_{\mathbf{p}}(\mathbb{R}^n)$, independent of the choice of α. It is spanned by the vectors $(\mathbf{p}; D\alpha(\mathbf{x}) \cdot \mathbf{e}_j) = (\mathbf{p}; \partial\alpha/\partial x_j)$, $j = 1, \ldots, k$, which are independent since $D\alpha$ has rank k; so they form a basis. The union of the $\mathcal{T}_{\mathbf{p}}(M)$ is the **tangent bundle** $\mathcal{T}(M)$; a **tangent vector field to** M is a continuous $F : M \to \mathcal{T}(M)$ with $F(\mathbf{p}) \in \mathcal{T}_{\mathbf{p}}(M)$.

29.2 Tensor fields

Definition 29.6 (tensor field, differential form)

Let A be open in $\mathbb{R}^n$. A **k-tensor field** in A is a function ω assigning to each $\mathbf{x} \in A$ a k-tensor $\omega(\mathbf{x}) \in \mathcal{L}^k(\mathcal{T}_{\mathbf{x}}(\mathbb{R}^n))$, such that $\omega(\mathbf{x})((\mathbf{x}; \mathbf{v}_1), \ldots, (\mathbf{x}; \mathbf{v}_k))$ is continuous as a function of $(\mathbf{x}, \mathbf{v}_1, \ldots, \mathbf{v}_k)$; if it is of class C^r, ω is a tensor field of class C^r. If $\omega(\mathbf{x})$ is alternating for each $\mathbf{x}$, ω is a **differential form** (or simply a **form**) of order k on A. More generally, a k-tensor field on an m-manifold M assigns to each $\mathbf{p} \in M$ an element of $\mathcal{L}^k(\mathcal{T}_{\mathbf{p}}(M))$; if alternating, it is a differential form on M.

A tensor field on an open set of $\mathbb{R}^n$ containing M restricts to one on M; conversely any tensor field on M extends to an open set containing M, but the proof is non-trivial. We restrict ourselves to tensor fields defined on open sets of $\mathbb{R}^n$.

Definition 29.7 (elementary forms)

$(\mathbf{x}; \mathbf{e}_1), \ldots, (\mathbf{x}; \mathbf{e}_n)$ is the **usual basis** of $\mathcal{T}_{\mathbf{x}}(\mathbb{R}^n)$. The **elementary 1-forms** $\tilde{\phi}_i$ on $\mathbb{R}^n$ are defined by $\tilde{\phi}_i(\mathbf{x})(\mathbf{x}; \mathbf{e}_j) = 0$ if $i \neq j$ and 1 if $i = j$. For an ascending k-tuple $I = (i_1, \ldots, i_k)$, the **elementary k-form** is $\tilde{\psi}_I(\mathbf{x}) = \tilde{\phi}_{i_1}(\mathbf{x}) \wedge \cdots \wedge \tilde{\phi}_{i_k}(\mathbf{x})$.

For each $\mathbf{x}$, $\tilde{\phi}_1(\mathbf{x}), \ldots, \tilde{\phi}_n(\mathbf{x})$ is the basis of $\mathcal{L}^1(\mathcal{T}_{\mathbf{x}}(\mathbb{R}^n))$ dual to the usual basis, and $\tilde{\psi}_I(\mathbf{x})$ is the corresponding elementary alternating tensor. They are of class C^∞, since

$$\tilde{\phi}_i(\mathbf{x})(\mathbf{x}; \mathbf{v}) = v_i, \qquad \tilde{\psi}_I(\mathbf{x})((\mathbf{x}; \mathbf{v}_1), \ldots, (\mathbf{x}; \mathbf{v}_k)) = \det X_I, \quad X = [\mathbf{v}_1 \ \cdots \ \mathbf{v}_k].$$

A k-form on A can be written uniquely as $\omega(\mathbf{x}) = \sum_{[I]} b_I(\mathbf{x})\tilde{\psi}_I(\mathbf{x})$; the scalar functions b_I are the **components** of ω relative to the standard elementary forms.

Lemma 29.2

Let ω be a k-form on the open set A of $\mathbb{R}^n$. Then ω is of class C^r if and only if its component functions b_I are of class C^r on A.

Proof. Write $\omega = \sum_{[I]} b_I \tilde{\psi}_I$. The $\tilde{\psi}_I$ are C^∞, so if the b_I are C^r, so is ω. Conversely, if ω is C^r as a function of $(\mathbf{x}, \mathbf{v}_1, \dots, \mathbf{v}_k)$, then for ascending J the function $\omega(\mathbf{x})((\mathbf{x};\mathbf{e}_{j_1}), \dots, (\mathbf{x};\mathbf{e}_{j_k}))$ is C^r in $\mathbf{x}$; it equals $b_J(\mathbf{x})$. ■

Lemma 29.3

Let ω and η be k-forms, and let θ be an l-form, on the open set A of $\mathbb{R}^n$. If ω and η and θ are of class C^r, so are $a\omega + b\eta$ and $\omega \wedge \theta$.

Proof. $a\omega + b\eta$ is a linear combination of C^r functions. Write $\omega = \sum_{[I]} b_I \tilde{\psi}_I$, $\theta = \sum_{[J]} c_J \tilde{\psi}_J$; then $\omega \wedge \theta = \sum_{[I]} \sum_{[J]} b_I c_J \tilde{\psi}_I \wedge \tilde{\psi}_J$. To express this in elementary forms, drop terms with repeated indices, rewrite the rest with indices in ascending order, and collect terms: each component of $\omega \wedge \theta$ is a sum, with signs ± 1, of functions $b_I c_J$, hence of class C^r (Lemma 29.2). ■

29.3 Differential forms of order zero

Definition 29.8 (scalar field; 0-form)

If A is open in $\mathbb{R}^n$ and $f : A \to \mathbb{R}$ is of class C^r, f is a **scalar field** in A, also called a **differential form of order** 0. The wedge product of 0-forms is $f \wedge g = f \cdot g$, and of a 0-form with a k-form is $(\omega \wedge f)(\mathbf{x}) = (f \wedge \omega)(\mathbf{x}) = f(\mathbf{x}) \cdot \omega(\mathbf{x})$.

All formal algebraic properties of the wedge product hold; anticommutativity holds since 0-forms have order 0: $f \wedge g = (-1)^0 g \wedge f$, $f \wedge \omega = (-1)^0 \omega \wedge f$.

Convention

Henceforth, Roman letters such as f, g, h denote 0-forms, and Greek letters such as ω, η, θ denote k-forms for $k > 0$.

§30. The Differential Operator

Motivation. The exterior derivative d unifies gradient, curl and divergence in one operator, and $d^2 = 0$ contains the identities $\operatorname{curl}\operatorname{grad} = 0$ and $\operatorname{div}\operatorname{curl} = 0$. It is what appears on the left side of Stokes' theorem.

The operator d carries k-forms to $(k+1)$-forms. We first define it on 0-forms.

30.1 The differential of a 0-form

Definition 30.1 (differential of a function)

Let A be open in $\mathbb{R}^n$ and $f : A \to \mathbb{R}$ of class C^r. The 1-form df on A is

$$df(\mathbf{x})(\mathbf{x};\mathbf{v}) = f'(\mathbf{x};\mathbf{v}) = Df(\mathbf{x}) \cdot \mathbf{v}.$$

It is the **differential** of f; it is of class C^{r-1} as a function of $\mathbf{x}$ and $\mathbf{v}$.

Theorem 30.1

The operator d is linear on 0-forms.

Proof. If $h = af+bg$, then $Dh(\mathbf{x}) = aDf(\mathbf{x})+bDg(\mathbf{x})$, so $dh(\mathbf{x})(\mathbf{x};\mathbf{v}) = a\,df(\mathbf{x})(\mathbf{x};\mathbf{v})+b\,dg(\mathbf{x})(\mathbf{x};\mathbf{v})$. ■

Lemma 30.2

Let $\tilde{\phi}_1,\dots,\tilde{\phi}_n$ be the elementary 1-forms in $\mathbb{R}^n$. Let $\pi_i : \mathbb{R}^n \to \mathbb{R}$ be the ith projection function, defined by the equation $\pi_i(x_1,\dots,x_n) = x_i$. Then $d\pi_i = \tilde{\phi}_i$.

Proof. $d\pi_i$ is a C^∞ 1-form, and $d\pi_i(\mathbf{x})(\mathbf{x};\mathbf{v}) = D\pi_i(\mathbf{x})\cdot\mathbf{v} = [0\cdots0\ 1\ 0\cdots0]\cdot\mathbf{v} = v_i$. ■

Convention

If $\mathbf{x}$ denotes the general point of $\mathbb{R}^n$, we denote the ith projection function $\mathbb{R}^n \to \mathbb{R}$ by x_i. Then dx_i equals the elementary 1-form $\tilde{\phi}_i$ in $\mathbb{R}^n$. If $I = (i_1,\dots,i_k)$ is an ascending k-tuple, we write $dx_I = dx_{i_1}\wedge\cdots\wedge dx_{i_k}$ for the elementary k-form $\tilde{\psi}_I$. The general k-form is uniquely $\omega = \sum_{[I]} b_I\,dx_I$. For an arbitrary k-tuple J we also set $dx_J = dx_{j_1}\wedge\cdots\wedge dx_{j_k}$; note dx_J is not the differential of a form but a wedge product of elementary 1-forms.

These forms are characterized by

$$dx_i(\mathbf{x})(\mathbf{x};\mathbf{v}) = v_i, \qquad dx_I(\mathbf{x})((\mathbf{x};\mathbf{v}_1),\dots,(\mathbf{x};\mathbf{v}_k)) = \det X_I, \quad X = [\mathbf{v}_1\cdots\mathbf{v}_k].$$

(Using x_i for the function π_i confuses a function with its value—an abuse as common as speaking of "the function x^3+2x+1".)

Theorem 30.3

Let A be open in $\mathbb{R}^n$; let $f : A \to \mathbb{R}$ be of class C^r. Then

$$df = (D_1 f)\,dx_1 + \cdots + (D_n f)\,dx_n.$$

In particular, $df = 0$ if f is a constant function.

In Leibniz notation, $df = \frac{\partial f}{\partial x_1}dx_1 + \cdots + \frac{\partial f}{\partial x_n}dx_n$—a formula in calculus books whose meaning is not explained there.

Proof. Evaluate both sides on $(\mathbf{x};\mathbf{v})$: $df(\mathbf{x})(\mathbf{x};\mathbf{v}) = Df(\mathbf{x})\cdot\mathbf{v}$, while $\sum_i D_i f(\mathbf{x})\,dx_i(\mathbf{x})(\mathbf{x};\mathbf{v}) = \sum_i D_i f(\mathbf{x})v_i$. ■

Convention

Since df is only C^{r-1} when f is C^r, to avoid counting degrees of differentiability we henceforth restrict ourselves to manifolds, maps, vector fields, and forms that are of class C^∞.

30.2 The differential of a k-form

Definition 30.2 ($\Omega^k(A)$)

For A open in $\mathbb{R}^n$, $\Omega^k(A)$ is the vector space of all (C^∞) k-forms on A, the **linear space of k-forms** on A.

Theorem 30.4

Let A be an open set in $\mathbb{R}^n$. There exists a unique linear transformation

$$d : \Omega^k(A) \to \Omega^{k+1}(A),$$

defined for $k \geq 0$, such that:

(1) If f is a 0-form, then df is the 1-form $df(\mathbf{x})(\mathbf{x};\mathbf{v}) = Df(\mathbf{x})\cdot\mathbf{v}$.

(2) If ω and η are forms of orders k and l, respectively, then $d(\omega\wedge\eta) = d\omega\wedge\eta + (-1)^k\omega\wedge d\eta$.

(3) For every form ω, $d(d\omega) = 0$.

d is the **differential operator**, and $d\omega$ the **differential** of ω.

Proof.

Step 1. Uniqueness. First, (2) and (3) imply $d(d\omega_1\wedge\cdots\wedge d\omega_k) = 0$ for any forms $\omega_1,\dots,\omega_k$: for $k=1$ this is (3); assuming it for $k-1$ and setting $\eta = d\omega_2\wedge\cdots\wedge d\omega_k$, (2) gives $d(d\omega_1\wedge\eta) = d(d\omega_1)\wedge\eta \pm d\omega_1\wedge d\eta$, both terms vanishing. Since d is linear, it suffices to compute d on $\omega = f\,dx_I$:

$$d\omega = d(f\wedge dx_I) = df\wedge dx_I + f\wedge d(dx_I) = df\wedge dx_I,$$

since $dx_I = dx_{i_1}\wedge\cdots\wedge dx_{i_k}$ with $dx_{i_j} = d(x_{i_j})$. So $d\omega$ is determined by d on 0-forms, which is (1).

Step 2. Definition. For $\omega = \sum_{[I]} f_I\,dx_I$ define $d\omega = \sum_{[I]} df_I\wedge dx_I$. It is C^∞: $d\omega = \sum_{[I]}[\sum_{j=1}^n (D_jf_I)dx_j]\wedge dx_I$; deleting terms with j among the indices of I, rearranging the rest into ascending order and collecting terms, each component of $d\omega$ is a linear combination of the D_jf_I. (If ω were only C^r, $d\omega$ would be C^{r-1}.) d is linear: for $\omega = \sum f_I dx_I$, $\eta = \sum g_I dx_I$, $d(a\omega+b\eta) = \sum d(af_I+bg_I)\wedge dx_I = \sum(a\,df_I + b\,dg_I)\wedge dx_I = a\,d\omega + b\,d\eta$.

Step 3. For an *arbitrary* k-tuple J, $d(f\,dx_J) = df\wedge dx_J$. If two indices of J coincide, $dx_J = 0$. Otherwise let I be the ascending rearrangement of J by the permutation π; by anticommutativity $dx_I = (\operatorname{sgn}\pi)dx_J$, and since d is linear and $\wedge$ homogeneous, $d(f\,dx_I) = df\wedge dx_I$ gives $(\operatorname{sgn}\pi)d(f\,dx_J) = (\operatorname{sgn}\pi)df\wedge dx_J$.

Step 4. Property (2) for $k=l=0$: $d(f\wedge g) = \sum_j D_j(f\cdot g)dx_j = \sum_j (D_jf)g\,dx_j + \sum_j f(D_jg)dx_j = df\wedge g + f\wedge dg$.

Step 5. Property (2) in general. For $k,l>0$, by linearity take $\omega = f\,dx_I$, $\eta = g\,dx_J$:

$$\begin{aligned} d(\omega\wedge\eta) &= d(fg\,dx_I\wedge dx_J) = d(fg)\wedge dx_I\wedge dx_J && \text{by Step 3,}\\ &= (df\,g + f\,dg)\wedge dx_I\wedge dx_J && \text{by Step 4,}\\ &= (df\wedge dx_I)\wedge(g\,dx_J) + (-1)^k(f\,dx_I)\wedge(dg\wedge dx_J) = d\omega\wedge\eta + (-1)^k\omega\wedge d\eta, \end{aligned}$$

the sign arising because dx_I is a k-form and dg a 1-form. If k or l is 0 the same argument works with dx_I or dx_J absent.

Step 6. If f is a 0-form, $d(df) = 0$:

$$d(df) = d\Big(\sum_j D_jf\,dx_j\Big) = \sum_j d(D_jf)\wedge dx_j = \sum_j\sum_i D_iD_jf\,dx_i\wedge dx_j = \sum_{i<j}(D_iD_jf - D_jD_if)\,dx_i\wedge dx_j,$$

which vanishes by equality of mixed partials (Theorem 6.3).

Step 7. If ω is a k-form, $k > 0$, $d(d\omega) = 0$. By linearity take $\omega = f\,dx_I$: $d(d\omega) = d(df \wedge dx_I) = d(df) \wedge dx_I - df \wedge d(dx_I)$ by (2). $d(df) = 0$ by Step 6, and $d(dx_I) = d(1) \wedge dx_I = 0$ by definition. ■

Definition 30.3 (closed and exact forms)

Let A be open in $\mathbb{R}^n$. A 0-form f on A is **exact** on A if it is constant on A; a k-form ω with $k > 0$ is **exact** on A if $\omega = d\theta$ for some $(k-1)$-form θ on A. A k-form ω $(k \geq 0)$ is **closed** on A if $d\omega = 0$.

Every exact form is closed: $df = 0$ for constant f, and $d(d\theta) = 0$. Conversely every closed form on A is exact if $A = \mathbb{R}^n$, or more generally if A is "star-convex" (Chapter 8), but not in general. If every closed k-form on A is exact, A is **homologically trivial in dimension** k.

Example 30.1

Let $A = \{(x, y) \in \mathbb{R}^2 : x \neq 0\}$ and $f(x, y) = x/|x|$. f is C^∞ on A and $df = 0$, but f is not exact on A since it is not constant.

Example 30.2

In differential equations, $P(x, y)\,dx + Q(x, y)\,dy = 0$ is called exact if there is f with $P = \partial f/\partial x$, $Q = \partial f/\partial y$; in our terms the 1-form $P\,dx + Q\,dy$ is the differential of f, hence exact. In $\mathbb{R}^3$, a vector field $\vec{F} = P\vec{i} + Q\vec{j} + R\vec{k}$ is *conservative* if it is the gradient of a scalar field f; this is the same as saying the form $P\,dx + Q\,dy + R\,dz$ is the differential of the 0-form f.

§31. Application to Vector and Scalar Fields

We show that the tensor-field formalism generalizes vector analysis in $\mathbb{R}^3$; these results are used in §38. $\Omega^k(A)$ and the set of C^∞ vector fields on A are linear spaces; we define linear maps translating between them.

Definition 31.1 (gradient, divergence)

Let A be open in $\mathbb{R}^n$, $f : A \to \mathbb{R}$ a scalar field. The **gradient** of f is the vector field

$$(\operatorname{grad} f)(\mathbf{x}) = (\mathbf{x}; D_1 f(\mathbf{x})\mathbf{e}_1 + \cdots + D_n f(\mathbf{x})\mathbf{e}_n).$$

If $G(\mathbf{x}) = (\mathbf{x}; g(\mathbf{x}))$ is a vector field with $g(\mathbf{x}) = g_1(\mathbf{x})\mathbf{e}_1 + \cdots + g_n(\mathbf{x})\mathbf{e}_n$, the **divergence** of G is the scalar field

$$(\operatorname{div} G)(\mathbf{x}) = D_1 g_1(\mathbf{x}) + \cdots + D_n g_n(\mathbf{x}).$$

Theorem 31.1

Let A be an open set in $\mathbb{R}^n$. There exist vector space isomorphisms α_i and β_j as in the following diagram:

$$\begin{array}{ccc} \text{Scalar fields in } A & \xrightarrow{\alpha_0} & \Omega^0(A) \\ {\scriptstyle \operatorname{grad}}\downarrow & & \downarrow{\scriptstyle d} \\ \text{Vector fields in } A & \xrightarrow{\alpha_1} & \Omega^1(A) \end{array} \qquad \begin{array}{ccc} \text{Vector fields in } A & \xrightarrow{\beta_{n-1}} & \Omega^{n-1}(A) \\ {\scriptstyle \operatorname{div}}\downarrow & & \downarrow{\scriptstyle d} \\ \text{Scalar fields in } A & \xrightarrow{\beta_n} & \Omega^n(A) \end{array}$$

such that $d \circ \alpha_0 = \alpha_1 \circ \operatorname{grad}$ and $d \circ \beta_{n-1} = \beta_n \circ \operatorname{div}$.

Proof. For scalar fields f, h and vector fields $F(\mathbf{x}) = (\mathbf{x}; \sum f_i(\mathbf{x})\mathbf{e}_i)$, $G(\mathbf{x}) = (\mathbf{x}; \sum g_i(\mathbf{x})\mathbf{e}_i)$ define

$$\alpha_0 f = f, \quad \alpha_1 F = \sum_{i=1}^{n} f_i\, dx_i, \quad \beta_{n-1} G = \sum_{i=1}^{n} (-1)^{i-1} g_i\, dx_1 \wedge \cdots \wedge \widehat{dx_i} \wedge \cdots \wedge dx_n, \quad \beta_n h = h\, dx_1 \wedge \cdots \wedge dx_n,$$

where $\widehat{a}$ means a is omitted. [Supplement]Each is linear, and bijective because $\{dx_i\}$, $\{dx_1 \wedge \cdots \wedge \widehat{dx_i} \wedge \cdots \wedge dx_n\}$ and $\{dx_1 \wedge \cdots \wedge dx_n\}$ are the elementary bases of Ω^1, Ω^{n-1}, Ω^n (Lemma 29.2). By Theorem 30.3, $d(\alpha_0 f) = df = \sum_i D_i f\, dx_i = \alpha_1(\operatorname{grad} f)$. And

$$d(\beta_{n-1} G) = \sum_i (-1)^{i-1} \sum_j D_j g_i\, dx_j \wedge dx_1 \wedge \cdots \wedge \widehat{dx_i} \wedge \cdots \wedge dx_n = \sum_i (-1)^{i-1} D_i g_i (-1)^{i-1} dx_1 \wedge \cdots \wedge dx_n = \beta_n(\operatorname{div} G),$$

since only $j = i$ survives and moving dx_i past $i-1$ factors gives the sign $(-1)^{i-1}$. ∎

For $\mathbb{R}^3$ there is also a *curl* operator.

Definition 31.2 (curl)

Let A be open in $\mathbb{R}^3$ and $F(\mathbf{x}) = (\mathbf{x}; \sum f_i(\mathbf{x})\mathbf{e}_i)$ a vector field. The **curl** of F is

$$(\operatorname{curl} F)(\mathbf{x}) = (\mathbf{x}; (D_2 f_3 - D_3 f_2)\mathbf{e}_1 + (D_3 f_1 - D_1 f_3)\mathbf{e}_2 + (D_1 f_2 - D_2 f_1)\mathbf{e}_3),$$

remembered as the symbolic determinant $\det \begin{bmatrix} \mathbf{e}_1 & \mathbf{e}_2 & \mathbf{e}_3 \\ D_1 & D_2 & D_3 \\ f_1 & f_2 & f_3 \end{bmatrix}$.

Theorem 31.2

Let A be an open set in $\mathbb{R}^3$. There exist vector space isomorphisms α_i and β_j as in the following diagram:

$$\begin{array}{ccc} \text{Scalar fields in } A & \xrightarrow{\alpha_0} & \Omega^0(A) \\ \downarrow{\scriptstyle \operatorname{grad}} & & \downarrow{\scriptstyle d} \\ \text{Vector fields in } A & \xrightarrow{\alpha_1} & \Omega^1(A) \\ \downarrow{\scriptstyle \operatorname{curl}} & & \downarrow{\scriptstyle d} \\ \text{Vector fields in } A & \xrightarrow{\beta_2} & \Omega^2(A) \\ \downarrow{\scriptstyle \operatorname{div}} & & \downarrow{\scriptstyle d} \\ \text{Scalar fields in } A & \xrightarrow{\beta_3} & \Omega^3(A) \end{array}$$

such that $d \circ \alpha_0 = \alpha_1 \circ \operatorname{grad}$, $d \circ \alpha_1 = \beta_2 \circ \operatorname{curl}$, and $d \circ \beta_2 = \beta_3 \circ \operatorname{div}$.

Proof. The maps are those of Theorem 31.1 with $n = 3$: $\beta_2 G = g_1\, dx_2 \wedge dx_3 - g_2\, dx_1 \wedge dx_3 + g_3\, dx_1 \wedge dx_2$. Only the middle equation needs checking [Supplement]:

$$\begin{aligned} d(\alpha_1 F) &= d(f_1 dx_1 + f_2 dx_2 + f_3 dx_3) = \sum_{i,j} D_j f_i\, dx_j \wedge dx_i \\ &= (D_2 f_3 - D_3 f_2)\, dx_2 \wedge dx_3 + (D_3 f_1 - D_1 f_3)\, dx_3 \wedge dx_1 + (D_1 f_2 - D_2 f_1)\, dx_1 \wedge dx_2, \end{aligned}$$

and since $dx_3 \wedge dx_1 = -dx_1 \wedge dx_3$, this is $\beta_2(\operatorname{curl} F)$. ∎

Consequently, $d(d\omega) = 0$ translates into $\operatorname{curl} \operatorname{grad} f = 0$ and $\operatorname{div} \operatorname{curl} F = 0$ in $\mathbb{R}^3$.

§32. The Action of a Differentiable Map

A C^∞ map $\alpha : A \to \mathbb{R}^n$, A open in $\mathbb{R}^k$, induces α_* on tangent spaces, and linear maps T induce dual maps T^* on alternating tensors. Combining these, α induces a dual transformation α^* of forms, preserving the vector space structure, the wedge product and d.

Definition 32.1 (pullback α^*)

Let A be open in $\mathbb{R}^k$, $\alpha : A \to \mathbb{R}^m$ of class C^∞, and B open in $\mathbb{R}^m$ containing $\alpha(A)$. The **dual transformation of forms** $\alpha^* : \Omega^l(B) \to \Omega^l(A)$ is defined as follows: for a 0-form f on B, $(\alpha^* f)(\mathbf{x}) = f(\alpha(\mathbf{x}))$; for an l-form ω on B, $l > 0$,

$$(\alpha^*\omega)(\mathbf{x})((\mathbf{x};\mathbf{v}_1),\dots,(\mathbf{x};\mathbf{v}_l)) = \omega(\alpha(\mathbf{x}))(\alpha_*(\mathbf{x};\mathbf{v}_1),\dots,\alpha_*(\mathbf{x};\mathbf{v}_l)).$$

Since $f, \omega, \alpha, D\alpha$ are C^∞, so are $\alpha^* f$ and $\alpha^*\omega$. If α is constant, $\alpha^* f$ is constant and $\alpha^*\omega$ is the zero tensor. If $\alpha(\mathbf{x}) = \mathbf{y}$ and $T = \alpha_* : \mathcal{T}_{\mathbf{x}}(\mathbb{R}^k) \to \mathcal{T}_{\mathbf{y}}(\mathbb{R}^m)$, then $T^*(\omega(\mathbf{y})) = (\alpha^*\omega)(\mathbf{x})$, since

$$T^*(\omega(\mathbf{y}))((\mathbf{x};\mathbf{v}_1),\dots,(\mathbf{x};\mathbf{v}_l)) = \omega(\alpha(\mathbf{x}))(\alpha_*(\mathbf{x};\mathbf{v}_1),\dots,\alpha_*(\mathbf{x};\mathbf{v}_l)).$$

So earlier results on dual transformations become results about forms:

Theorem 32.1

Let A be open in $\mathbb{R}^k$; let $\alpha : A \to \mathbb{R}^m$ be a C^∞ map. Let B be open in $\mathbb{R}^m$ and contain $\alpha(A)$; let $\beta : B \to \mathbb{R}^n$ be a C^∞ map. Let ω, η, θ be forms defined in an open set C of $\mathbb{R}^n$ containing $\beta(B)$; assume ω and η have the same order. The transformations α^* and β^* have the following properties:
(1) $\beta^*(a\omega + b\eta) = a(\beta^*\omega) + b(\beta^*\eta)$.
(2) $\beta^*(\omega \wedge \theta) = \beta^*\omega \wedge \beta^*\theta$.
(3) $(\beta \circ \alpha)^*\omega = \alpha^*(\beta^*\omega)$.

Proof. For forms of positive order, (1) and (3) restate Theorem 26.5 (using Lemma 29.1 for (3)), and (2) restates property (6) of Theorem 28.1. [Supplement](Order zero.) If ω, η are 0-forms, $\beta^*(af+bg) = (af+bg)\circ\beta = a(f\circ\beta)+b(g\circ\beta)$. If $\theta = g$ is a 0-form, $\beta^*(\omega\wedge g)(\mathbf{y}) = g(\beta(\mathbf{y}))\cdot\beta^*\omega(\mathbf{y}) = (\beta^*\omega \wedge \beta^* g)(\mathbf{y})$, and similarly if ω is a 0-form. For 0-forms, (3) reads $f \circ (\beta \circ \alpha) = (f \circ \beta) \circ \alpha$. ■

We now compute $\alpha^*\omega$ when ω is a 1-form and when ω is a k-form on $\mathbb{R}^n$ with $\alpha : A \to \mathbb{R}^n$, A open in $\mathbb{R}^k$; since α^* is linear, preserves wedge products and $\alpha^* f = f \circ \alpha$, it suffices to treat elementary forms.

Theorem 32.2

Let A be open in $\mathbb{R}^k$; let $\alpha : A \to \mathbb{R}^n$ be a C^∞ map. Let $\mathbf{x}$ denote the general point of $\mathbb{R}^k$; let $\mathbf{y}$ denote the general point of $\mathbb{R}^n$. Then dx_i and dy_i denote the elementary 1-forms in $\mathbb{R}^k$ and $\mathbb{R}^n$, respectively.
(a) $\alpha^*(dy_i) = d\alpha_i$.
(b) If $I = (i_1,\dots,i_k)$ is an ascending k-tuple from the set $\{1,\dots,n\}$, then

$$\alpha^*(dy_I) = \left(\det \frac{\partial \alpha_I}{\partial \mathbf{x}}\right) dx_1 \wedge \cdots \wedge dx_k, \qquad \text{where} \quad \frac{\partial \alpha_I}{\partial \mathbf{x}} = \frac{\partial(\alpha_{i_1},\dots,\alpha_{i_k})}{\partial(x_1,\dots,x_k)}.$$

Proof. (a) With $\mathbf{y} = \alpha(\mathbf{x})$,

$$(\alpha^*(dy_i))(\mathbf{x})(\mathbf{x};\mathbf{v}) = dy_i(\mathbf{y})(\alpha_*(\mathbf{x};\mathbf{v})) = i\text{th component of } D\alpha(\mathbf{x})\cdot\mathbf{v} = \sum_{j=1}^k D_j\alpha_i(\mathbf{x})v_j = \sum_{j=1}^k \frac{\partial \alpha_i}{\partial x_j}(\mathbf{x})\, dx_j(\mathbf{x})(\mathbf{x};\mathbf{v}).$$

So $\alpha^*(dy_i) = \sum_j \frac{\partial \alpha_i}{\partial x_j} dx_j$, which equals $d\alpha_i$ by Theorem 30.3.

(b) $\alpha^*(dy_I)$ is a k-form on an open set of $\mathbb{R}^k$, so $\alpha^*(dy_I) = h\, dx_1 \wedge \cdots \wedge dx_k$ for a scalar function h, found by evaluating on $((\mathbf{x};\mathbf{e}_1),\dots,(\mathbf{x};\mathbf{e}_k))$:

$$h(\mathbf{x}) = dy_I(\mathbf{y})(\alpha_*(\mathbf{x};\mathbf{e}_1),\dots,\alpha_*(\mathbf{x};\mathbf{e}_k)) = dy_I(\mathbf{y})((\mathbf{y};\partial\alpha/\partial x_1),\dots,(\mathbf{y};\partial\alpha/\partial x_k)) = \det[D\alpha(\mathbf{x})]_I = \det\frac{\partial \alpha_I}{\partial \mathbf{x}}.$$

■

Formula (a) is easy to remember: to compute $\alpha^*(dy_i)$, take dy_i and substitute $y_i = \alpha_i(\mathbf{x})$. One could compute $\alpha^*(dy_I) = d\alpha_{i_1} \wedge \cdots \wedge d\alpha_{i_k}$, but the wedge product is laborious if $k > 2$.

Theorem 32.3

Let A be open in $\mathbb{R}^k$; let $\alpha : A \to \mathbb{R}^n$ be of class C^∞. If ω is an l-form defined in an open set of $\mathbb{R}^n$ containing $\alpha(A)$, then

$$\alpha^*(d\omega) = d(\alpha^*\omega).$$

Proof. Let $\mathbf{x}$, $\mathbf{y}$ be the general points of $\mathbb{R}^k$, $\mathbb{R}^n$.

Step 1. For a 0-form f:

$$\alpha^*(df) = \alpha^*\Big(\sum_{i=1}^n (D_i f)\, dy_i\Big) = \sum_{i=1}^n ((D_i f) \circ \alpha)\, d\alpha_i. \tag{$*$}$$

On the other hand

$$d(\alpha^* f) = d(f \circ \alpha) = \sum_{j=1}^k D_j(f \circ \alpha)\, dx_j. \tag{$**$}$$

By the chain rule, $D(f \circ \alpha)(\mathbf{x}) = Df(\mathbf{y}) \cdot D\alpha(\mathbf{x})$, so $D_j(f \circ \alpha) = \sum_i ((D_i f) \circ \alpha) \cdot D_j \alpha_i$. Substituting in $(**)$,

$$d(\alpha^* f) = \sum_{j=1}^k \sum_{i=1}^n ((D_i f) \circ \alpha) D_j \alpha_i\, dx_j = \sum_{i=1}^n ((D_i f) \circ \alpha)\, d\alpha_i, \tag{$***$}$$

which agrees with $(*)$.

Step 2. For forms of positive order, by linearity it suffices to take $\omega = f\, dy_I$ with I ascending. Then

$$\alpha^*(d\omega) = \alpha^*(df \wedge dy_I) = \alpha^*(df) \wedge \alpha^*(dy_I), \tag{$\dagger$}$$

$$d(\alpha^*\omega) = d[(\alpha^* f) \wedge \alpha^*(dy_I)] = d(\alpha^* f) \wedge \alpha^*(dy_I) + (\alpha^* f) \wedge 0, \tag{$\dagger\dagger$}$$

since $d(\alpha^*(dy_I)) = d(d\alpha_{i_1} \wedge \cdots \wedge d\alpha_{i_l}) = 0$. Compare $(\dagger)$ and $(\dagger\dagger)$ using Step 1. ■

With this, the algebra of differential forms and d, as summarized in §30 and this section, is all we need.

Forms versus vector fields

Here forms appear more natural than vector fields. A C^∞ map $\alpha : A \to \mathbb{R}^n$ induces α_* on tangent vectors, but there is no way to carry a vector *field* on A to one on $\alpha(A)$: if $\alpha(\mathbf{x}_1) = \alpha(\mathbf{x}_2) = \mathbf{y}$ with $\mathbf{x}_1 \neq \mathbf{x}_2$, a field $F(\mathbf{x}) = (\mathbf{x}; f(\mathbf{x}))$ gives two possibly different vectors $\alpha_*(\mathbf{x}_1; f(\mathbf{x}_1))$, $\alpha_*(\mathbf{x}_2; f(\mathbf{x}_2))$ at $\mathbf{y}$. For a diffeomorphism $\alpha : A \to B$ one can define $\tilde{\alpha}_* F = G$ by $G(\mathbf{y}) = \alpha_*(F(\alpha^{-1}(\mathbf{y})))$, and $\tilde{\alpha}_* h = h \circ \alpha^{-1}$ for scalar fields; but this does not in general commute with grad, curl, div, nor with the translation operators α_i, β_j of §31. Forms pull back under *any* C^∞ map, compatibly with $\wedge$ and d.